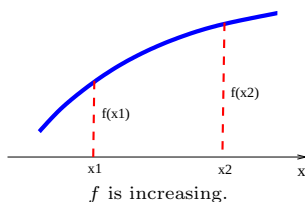
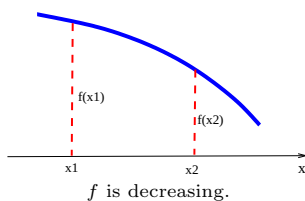


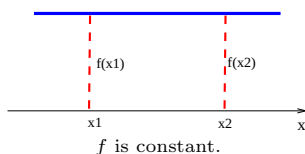
Increasing Function. A function f is said to be *increasing* on an interval if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in that interval.



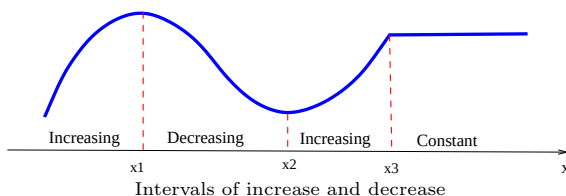
Decreasing Function. A function f is said to be *decreasing* on an interval if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in that interval.



Constant Function. A function f is said to be *constant* on an interval if $f(x_1) = f(x_2)$ for all x_1, x_2 in that interval.



Remark If a function is given in terms of its graph we can find the intervals on which the function remains increasing, decreasing or constant, as you can see in the following figure.



However, if a function is given in terms of an algebraic expression, we use the following result to find such intervals.

First Derivative Test. Let f be continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

- If $f'(x) > 0$ for every x in (a, b) , then f is increasing on $[a, b]$.
- If $f'(x) < 0$ for every x in (a, b) , then f is decreasing on $[a, b]$.
- If $f'(x) = 0$ for every x in (a, b) , then f is constant on $[a, b]$.

Example 1. Find the intervals on which $f(x) = x^2 - 4x + 3$ is increasing and the intervals on which it is decreasing.

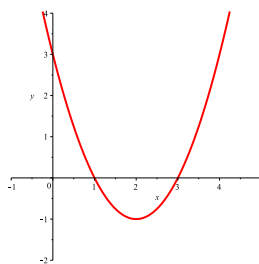
Solution.

Step 1. [Derivative and Its Factorization] The derivative of f is $f'(x) = 2x - 4$.

Step 2. [Intervals and Sign Analysis] The solution of the equation $2x - 4 = 0$ is $x = 2$. This point ($x = 2$) divides the number line into two intervals, $(-\infty, 2]$, and $[2, +\infty)$.

Interval	Test Point	Sign of $f'(x)$	Increasing/Decreasing
$(-\infty, 2]$	0	—	Decreasing
$[2, +\infty)$	3	+	Increasing

Step 3. [Sketch]



—Solution Ends—

Example 2. Find the intervals on which $f(x) = x^3$ is increasing and the intervals on which it is decreasing.

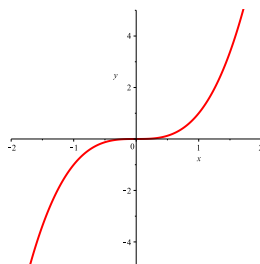
Solution.

Step 1. [Derivative and Its Factorization] The derivative of f is $f'(x) = 3x^2$.

Step 2. [Intervals and Sign Analysis] The solution of the equation $3x^2 = 0$ is $x = 0$. This point ($x = 0$) divides the number line into two intervals, $(-\infty, 0]$, and $[0, +\infty)$.

Interval	Test Point	Sign of $f'(x)$	Increasing/Decreasing
$(-\infty, 0]$	-1	-	Decreasing
$[0, +\infty)$	1	+	Increasing

Step 3. [Sketch]



—Solution Ends—

Example 3. Find the intervals on which $f(x) = 5 + 12x - x^3$ is increasing and the intervals on which it is decreasing.

Solution.

Step 1. [Derivative and Its Factorization] The derivative of f along with its factorization is

$$f'(x) = 12 - 3x^2 = 3(4 - x^2) = 3(2 + x)(2 - x).$$

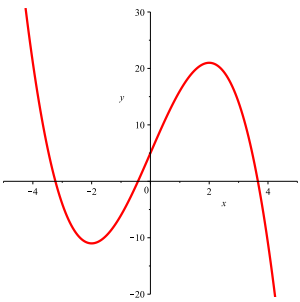
Step 2. [Intervals and Sign Analysis] Put each linear factor in $f'(x)$ equal to zero and solve for x . That is, solve the equations $2 + x = 0$ and $2 - x = 0$. Their solutions are $x = -2$ and $x = 2$. These points divide the number line into three intervals, $(-\infty, -2]$, $[-2, 2]$, and $[2, +\infty)$.

Interval	Test Point	Sign of $f'(x)$	Increasing/Decreasing
$(-\infty, -2]$	-3	-	Decreasing
$[-2, 2]$	0	+	Increasing
$[2, +\infty)$	3	-	Decreasing

Step 3. [Sketch]

—Solution Ends—

Example 4. Find the intervals on which $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$ is increasing and the intervals on which it is decreasing.



Solution.

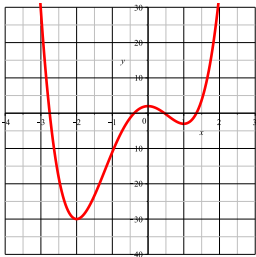
Step 1. [Derivative and Its Factorization] The derivative of f along with its factorization is

$$\begin{aligned} f'(x) &= 12x^3 + 12x^2 - 24x \\ &= 12x(x^2 + x - 2) \\ &= 12x(x^2 + 2x - x - 2) \\ &= 12x(x(x + 2) - (x + 2)) \\ &= 12x(x + 2)(x - 1) \end{aligned}$$

Step 2. [Intervals and Sign Analysis] Put each linear factor in $f'(x)$ equal to zero and solve for x . That is, solve the equations $12x = 0$, $x + 2 = 0$, and $x - 1 = 0$. Their solutions are $x = 0$, $x = -2$, and $x = 1$. These points divide the number line into four intervals, $(-\infty, -2)$, $(-2, 0)$, $(0, 1)$, and $(1, +\infty)$.

Interval	Test Point	Sign of $f'(x)$	Increasing/Decreasing
$(-\infty, -2]$	-3	$-$	Decreasing
$[-2, 0]$	-1	$+$	Increasing
$[0, 1]$	0.5	$-$	Decreasing
$[1, +\infty)$	2	$+$	Increasing

Step 3. [Sketch]



$$f(x) = 3x^4 + 4x^3 - 12x^2 + 2$$

Practice Problems.

Find the intervals on which f is increasing and the intervals on which it is decreasing.

1. $f(x) = x^3 - 3x^2 + 1$
2. $f(x) = 3x^4 - 4x^3$.
3. $f(x) = x^3 - x^2 - 2x$.

(I shall welcome your suggestions to improve these notes.)